

ABAKAN, Russian Federation

WMO: 287854

Lat: 53.740N

Lon: 91.385E

Elev: 831

StdP: 14.26

Time Zone: 7.00 (E07)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.6	-27.4	-38.6	0.6	-31.8	-33.1	0.9	-26.9	22.6	19.0	20.4	17.4	1.1	190	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	88.2	66.4	85.7	65.6	82.2	64.1	69.4	82.8	67.6	80.2	66.1	78.0	7.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.7	94.3	73.7	62.9	88.5	72.1	61.2	83.3	71.2	33.8	83.2	32.4	80.4	31.2	77.9	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.3	19.2	16.7	-33.7	94.0	7.1	3.9	-38.8	96.8	-43.0	99.0	-46.9	101.2	-52.1	104.0		
(5)			-33.8	74.0	7.8	4.0	-39.4	76.9	-44.0	79.2	-48.4	81.4	-54.0	84.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.6	-1.3	4.3	24.3	41.2	53.1	65.9	68.8	64.2	52.1	37.6	21.2	6.1		
	DBStd	26.38	15.20	15.76	12.19	9.08	8.81	6.56	5.48	5.98	7.47	8.64	12.11	14.36		
	HDD50	6703	1592	1278	795	285	71	1	0	0	61	394	865	1361		
	HDD65	10669	2057	1698	1260	713	378	68	24	87	392	851	1315	1826		
	CDD50	1823	0	0	0	22	167	478	583	441	124	8	0	0		
	CDD65	314	0	0	0	0	9	95	143	63	4	0	0	0		
	CDH74	3491	0	0	0	14	237	1097	1361	697	85	0	0	0		
	CDH80	1119	0	0	0	3	55	355	463	219	24	0	0	0		
(14) Wind	WSAvg	6.1	4.0	4.6	6.8	8.1	8.1	6.5	5.7	5.4	5.9	6.4	6.2	5.0		
(15) Precipitation	PrecAvg	13.0	0.3	0.2	0.2	0.6	1.2	2.4	2.7	2.5	1.4	0.7	0.5	0.4		
	PrecMax	20.9	0.6	0.8	0.9	1.3	2.6	6.4	5.2	5.4	3.2	2.1	1.5	0.8		
	PrecMin	9.3	0.0	0.0	0.1	0.0	0.4	0.9	0.9	0.9	0.2	0.2	0.1	0.2		
	PrecStd	2.3	0.2	0.2	0.2	0.3	0.6	1.1	0.9	1.2	0.6	0.4	0.3	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.7	39.1	57.1	75.2	85.8	91.5	93.6	91.4	84.0	66.2	49.8	37.3		
		MCWB	34.5	33.1	43.3	53.0	61.4	65.4	68.5	68.9	62.3	51.5	42.3	32.9		
	2%	DB	31.8	34.2	50.1	68.4	80.6	87.7	89.3	85.8	75.1	59.4	44.8	32.4		
		MCWB	28.0	29.8	40.6	50.7	58.4	65.6	67.0	67.4	59.8	49.2	38.5	28.5		
	5%	DB	26.6	30.6	44.7	63.0	75.4	84.2	85.8	82.0	69.8	55.2	41.0	30.0		
		MCWB	23.7	27.3	37.0	48.3	56.5	64.1	66.4	65.3	57.7	46.5	36.1	26.8		
	10%	DB	21.2	26.7	40.8	58.6	70.2	80.6	82.1	77.2	65.8	50.4	37.2	26.6		
		MCWB	18.8	23.7	34.4	46.2	54.3	63.3	65.0	63.2	55.3	43.5	33.2	23.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.0	33.5	45.3	55.0	65.2	70.3	72.3	71.8	64.5	53.2	43.0	33.1		
		MCDB	37.9	37.6	55.6	72.0	82.4	84.8	87.9	86.5	79.9	63.7	47.6	36.4		
	2%	WB	27.9	30.3	40.8	52.1	61.0	68.0	69.7	68.5	60.9	49.9	39.4	29.2		
		MCDB	31.2	33.8	48.8	65.7	74.1	81.8	82.8	81.2	71.2	58.8	44.3	32.5		
	5%	WB	23.7	27.1	37.6	49.4	58.2	66.3	68.0	66.5	58.6	47.1	36.2	26.3		
		MCDB	26.6	30.5	44.1	62.0	72.0	79.4	80.6	78.0	67.8	53.8	40.5	29.6		
	10%	WB	18.7	23.8	34.5	46.4	55.7	64.5	66.7	64.5	56.4	44.3	32.9	23.7		
		MCDB	21.0	26.7	39.9	56.7	67.6	76.7	78.4	74.7	64.1	49.6	36.5	26.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.9	20.2	21.2	23.4	23.8	23.0	20.9	21.6	20.9	18.0	14.2	15.3		
		MCDBR	17.4	20.4	25.2	33.2	33.1	29.9	27.4	29.3	29.2	24.5	17.0	14.6		
		MCWBR	15.2	16.8	16.9	18.3	16.2	12.1	9.9	13.0	15.4	15.1	12.3	12.3		
		MCDBR	17.5	20.3	23.5	31.2	28.1	25.3	23.2	26.0	26.2	23.9	16.7	14.8		
	5% WB	MCWBR	15.5	17.0	16.4	18.2	15.2	11.4	9.8	12.6	14.3	15.3	12.6	12.8		
(40) Clear-Sky Solar Irradiance	taub		0.280	0.315	0.363	0.404	0.411	0.403	0.415	0.401	0.362	0.344	0.318	0.284		
	taud		2.065	2.021	1.999	2.151	2.189	2.317	2.304	2.314	2.358	2.284	2.128	2.017		
	Ebn at Noon		207	238	252	260	265	269	263	259	255	229	188	173		
	Edh at Noon		23	33	43	42	43	38	38	36	30	26	22	20		
(44) All-Sky Solar Radiation	RadAvg		317	646	1118	1429	1690	1931	1747	1466	1031	572	284	206		
	RadStd		23	44	90	76	163	159	109	112	95	51	31	20		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page